# Python Weather Application

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**Purpose**

The Python Weather Application provides users with current weather information for locations in the United States. The program interacts with the OpenWeatherMap API to retrieve real-time weather data using either a city/state combination or ZIP code. The application emphasizes input validation, user-friendly outputs with symbols, and robust error handling.

**Features**

1. **City/State Lookup** – Users can input a city name and state code to retrieve weather data.
2. **ZIP Code Lookup** – Users can provide a standard 5-digit or ZIP+4 format ZIP code.
3. **Unit Selection** – Temperature can be displayed in Celsius, Fahrenheit, or Kelvin.
4. **Real-Time Weather Details** – Includes temperature, feels like, high/low, pressure, humidity, cloud cover, wind speed, visibility, and description.
5. **Symbols and Formatting** – Uses Unicode symbols for weather elements to enhance readability.
6. **Error Handling** – Handles invalid inputs, API connection errors, and missing data gracefully.
7. **Validation** – Ensures ZIP codes and state codes exist in the United States.

**Modules and Libraries**

- requests – For HTTP requests to the OpenWeatherMap API.
- colorama – For colored console output.
- uszipcode – To validate ZIP codes.
- urllib.parse.quote – For safely encoding city/state names in URLs.
- re – Regular expressions for ZIP code validation.

**How It Works**

1. Users select either City/State or ZIP code lookup.
2. Input is validated and formatted appropriately.
3. The program fetches geographic coordinates using the OpenWeatherMap Geo API.
4. Using the coordinates, the program retrieves weather information from the OpenWeatherMap Weather API.
5. Weather data is displayed in a user-friendly format with symbols for better readability.
6. Users can exit the application gracefully.

**Error Handling**

- API HTTP errors, connection errors, and timeouts are handled with clear messages.
- Invalid ZIP codes, city names, or state codes prompt informative feedback.
- Missing or invalid temperature units are validated.

**Example Output**

📍 Location : New York, US
🌡️ Current Temperature : 15 °C
😊 Feels Like : 14 °C
🔽 Low Temperature : 12 °C
🔼 High Temperature : 17 °C
⚙️ Pressure : 1015 hPa
💧 Humidity : 60%
☁️ Cloud Cover : 40%
💨 Wind Speed : 3 m/s
👀 Visibility : 10000 meters
🌈 Description : scattered clouds

**Conclusion**

This Python program demonstrates a functional, interactive, and user-friendly weather application that leverages APIs for real-time data. Its robust input validation and error handling ensure a reliable user experience.